ISSUE NO 09 REPORT

HelpDesk Net-50K: Noisy Label Analysis

Executive Summary:

A fine tuned classifier trained on Helpdesk Net-50k (50k tickets, 6 classes) levels off at 825 accuracy. This report looks at whether the real issue is label noise rather than the mode.

Almost 15% of crowd labels are different or in conflict with the expert labels (95% CI 11.8% to 18.8%) i.e. approximately 13.5% across the whole dataset. Tier B workers are significantly noisier than Tier A (21.8% vs 6.7%, p = 2.4e-5). The errors are systematic and focused on a small number of overlapping class pairs (primarily Billing/Refund and Technical/Account Access). A simple uniform noise model suggests a true accuracy of around 96%, though this is quite fragile and likely an overestimate. Looking at 2000 tickets won't fix it all so it will need to be combined with noise-robust training and a clean expert labelled test set.

All the numbers are from the problem statement. There was no raw data available and the Milestone 4 results are from a simulation using an assumed model confidence signal.

Milestone 1: Quantify Label Noise

Method: the expert audit is a random sample of 400 crowd-labeled tickets. Experts are the ground truth so the noise rate is a proportion. I've used Wilson confidence intervals (better than Wald for small samples) and tested the tiers against each other using a two-proportion z-test, cross-checked with Fisher's exact test.

Group	Disagreements	Rate	Wilson 95% CI
Overall crowd	60 / 400	15.0%	[11.8%, 18.8%]
Tier A	12 / 180	6.7%	[3.9%, 11.3%]
Tier B	48 / 220	21.8%	[16.9%, 27.7%]
Full 50K dataset*	n/a	13.5%	[10.7%, 16.9%]

Assumes the 5000 expert tickets are clean (0.9 x 15%)

Tier comparison. Difference is 15.2 percentage points (95% CI [8.6, 21.7]). Z-test: z = 4.22, p = 2.4 x 10^-5. Fisher's exact: p = 1.8 x 10^-5, OR 3.91 (RR ~3.3). Very unlikely to be chance.

Caveats. Experts aren't perfect so disagreement is a proxy for noise. The tier samples are small so their intervals are wide. All of it is predicated on the audit sample being random.

Milestone 2: Characterize the Noise

Pattern. Only 8 of the 30 possible confusions occur. Three pairs account for 86.7% of all the disagreements: Billing/Refund (26), Technical/Account Access (15) and Shipping/Other (11). The majority of the confusions are in both directions which again points to fuzzy class boundaries rather than a one way bias. Technical → Other and Billing → Other have no reverse errors which suggests that 'Other' is being used as a catch all when the annotators aren't sure.

Test. A Monte Carlo simulation (10,000 runs) spreads 60 errors uniformly over 30 cells. Observed χ^2 is 222.0, simulated maximum is 64.0 ($p < 0.0001$). Non-zero cells are 8 observed vs about 26 expected. So the noise is not uniform. The null model is very simple (larger classes will of course produce more errors) so I treat the test as supporting evidence rather than proof.

Most likely root cause: overlapping/ambiguous class definitions combined with poor training of the newer staff (this is consistent with Tier B being considerably worse). This is of course just a hypothesis.

Process changes.

- Write rules for making decisions and give examples of boundaries for pairs that are easy to confuse.
- Make 'Other' a genuine last resort and always ask for a reason.
- Check per-worker confusion patterns.
- Add gold standard tickets to monitor quality.
- Collect multiple labels on the ambiguous tickets and measure the agreement (kappa).
- Consider a merge or a restructure of Billing and Refund.
- Retrain the Tier B workers or down-weight their labels.

Milestone 3: Accuracy Ceiling

Let a be the true accuracy, $\eta = 0.15$ and $K = 6$. A prediction can be correct in two scenarios: (i) the model is right and the label is unchanged, or (ii) the model is wrong and the label flips to the same wrong class (probability $\eta/(K-1) = 0.03$):

$0.82 = a(0.85) + (1 - a)(0.03)$. So, $a = 0.79 / 0.82$, which is about 0.963.

Sensitivity: $\eta = 0.118$ gives 92.8%, $\eta = 0.135$ gives 94.6%, $\eta = 0.15$ gives 96.3% and $\eta = 0.188$ gives 101% (impossible). A perfect model would score $1-\eta$ (so 82% at $\eta = 18\%$). It breaks above that point, which shows how fragile the formula is.

Why the estimate is unreliable.

- The noise depends on the class. It is not uniform.
- The model may have simply learned the annotators' consistent errors (which would inflate the observed accuracy) and as a result the true accuracy could be considerably lower.

- The noise rate varies by tier and is purely an estimate.
- Small classes such as 'Other' are the ones affected to a greater extent.
- Experts are imperfect too.

So, 96% is a very rough, optimistic figure rather than a real measurement.

Milestone 4: Find the Bad Labels

Selection strategy: rank the tickets by their estimated probability of being mislabelled using three observable signals (out-of-fold model confidence in the label, annotator tier and class pair risk). The selector must never use the unknown true class. Do approximately 1700 reviews on the top ranked tickets and approximately 300 (15%) on random tickets. The random slice provides unbiased estimates, exposes blind spots and provides data to fit the signal weights (e.g. via logistic regression).

Evaluate the selector. Look at precision (share of reviewed tickets that were wrong) versus the random baseline of circa 15%, lift (precision / baseline), recall (errors found out of all errors, capped around 30%) and the precision@k curves. Validate on a held out audited slice of the real data.

Simulation (illustrative). I constructed a synthetic world from the audit data (noise 14.6%, 71.8% of errors in systematic pairs). Model confidence is at three levels and the errors in the systematic pairs are presented as correct labels as models are prone to learn them.

Signal	Random	Priors only	Model only	Model + priors	Residual noise
Strong	14.8%	38.0%	71.2%	72.9%	11.4%
Moderate	16.0%	38.0%	34.0%	42.4%	12.8%
Weak	15.2%	38.0%	24.6%	35.9%	13.1%

Random review stays at 15%. Priors are most useful when the model signal is weak. Best case recall is only 22.1%. These figures are dependent on the assumed signal and a single seed so they're indicative of behaviour rather than actual performance.

Using corrected labels.

1. Replace the training labels on the reviewed tickets with the expert labels.
2. Estimate the noise transition matrix from the random slice (plus the original audit) as the selected tickets are biased towards errors.
3. Review in small batches, retrain and re-score.
4. Use the corrected labels as input for noise-robust training.
5. Keep the reviewed tickets out of the test set.

Residual noise. With 6,592 simulated errors the review process leaves about 11-13% noise. It's a complement to good training, not a replacement.

Milestone 5: Mitigate Noise and Evaluate Reliably

Two training-time methods.

1. **Forward loss correction.** Estimate a transition matrix T (probability of each crowd label given the true class) from the audit and review data. The model predicts the clean class and T then converts that into a noisy label prediction prior to computing the loss. It learns class-specific noise but errors in T are bad and the audit only has 60 errors.
2. **Generalised Cross-Entropy (GCE**). A bounded loss which prevents the model from fitting obviously wrong labels too hard. Requires no transition matrix, just weaker on systematic noise.

Trustworthy evaluation set.

- Label it with the experts, not the crowd.
- Keep it separate from training and from selector tuning.
- Use a number of experts with adjudication on at least a subset.
- Make it representative with a good number of the rare class examples (around 250 'Other' tickets in the 5000).
- Report accuracy with confidence intervals, as well as macro-F1 and per-class recall. Report the hard pairs separately.

Why the crowd test set is a problem.

- Approximately 15% of the labels are incorrect which caps and biases the accuracy.
- A model which simply copies the annotators' systematic errors is rewarded, so we can't tell the difference between a better model and one that just copies the errors.
- Differences of a point or two get lost in the noise.
- Train and test share the same noise process so overfitting to noise remains invisible.

Limitations and Recommendations

Limitations. Experts are treated as truth. The audit is small. The root cause is likely, not proven. The simulation uses an assumed confidence signal and one seed. The Milestone 3 formula assumes uniform noise.

Recommendations.

1. Rewrite guidelines for confusable pairs, restrict "Other", add gold questions and multi-labeling.
2. Audit and retrain Tier B or down-weight their labels.
3. Use the 2000 reviews on a combined selector with around 15% random.
4. Train with forward loss correction and a strong loss.

5. Evaluate only on an expert labelled, disjoint, adjudicated test set.